

Topic 1: Atomic Structure and the Periodic Table

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include the use of hand-held spectrosopes to investigate spectra from flame tests.

Mathematical skills that could be developed in this topic include calculating a relative atomic mass from isotopic composition data, using simple probability to calculate the peak heights for the mass spectrum of chlorine molecules, using logarithms to compare successive ionisation energies for an element.

Within this topic, students can consider how models for the atom have developed over time, as new evidence has become available. They can also consider how data is used to investigate relationships, such as between the magnitude of ionisation energy and the structure of an atom.

Students should:

1. know the structure of an atom in terms of electrons, protons and neutrons
2. know the relative mass and relative charge of protons, neutrons and electrons
3. know what is meant by the terms 'atomic (proton) number' and 'mass number'
4. be able to determine the number of each type of sub-atomic particle in an atom, molecule or ion from the atomic (proton) number and mass number
5. understand the term 'isotopes'
6. be able to define the terms 'relative isotopic mass' and 'relative atomic mass', based on the ^{12}C scale
7. understand the terms 'relative molecular mass' and 'relative formula mass', including calculating these values from relative atomic masses
Definitions of these terms will not be expected.
The term 'relative formula mass' should be used for compounds with giant structures.
8. be able to analyse and interpret data from mass spectrometry to calculate relative atomic mass from relative abundance of isotopes and vice versa
9. be able to predict the mass spectra, including relative peak heights, for diatomic molecules, including chlorine
10. understand how mass spectrometry can be used to determine the relative molecular mass of a molecule
Limited to the m/z value for the molecular ion, M^+ , giving the relative molecular mass of the molecule.
11. be able to define the terms 'first ionisation energy' and 'successive ionisation energies'
12. understand how ionisation energies are influenced by the number of protons, the electron shielding and the electron sub-shell from which the electron is removed
13. understand reasons for the general increase in first ionisation energy across a period

Students should:

15. understand how ideas about electronic configuration developed from:
 - i the fact that atomic emission spectra provide evidence for the existence of quantum shells
 - ii the fact that successive ionisation energies provide evidence for the existence of quantum shells and the group to which the element belongs
 - iii the fact that the first ionisation energy of successive elements provides evidence for electron sub-shells
16. know the number of electrons that can fill the first four quantum shells
17. know that an orbital is a region within an atom that can hold up to two electrons with opposite spins
18. know the shape of an *s*-orbital and a *p*-orbital
19. know the number of electrons that occupy *s*, *p* and *d*-subshells
20. know that electrons fill subshells singly, before pairing up, and that two electrons in the same orbital must have opposite spins
21. be able to predict the electronic configurations, using 1s notation and electrons-in-boxes notation, of:
 - i atoms, given the atomic number, *Z*, up to $Z = 36$
 - ii ions, given the atomic number, *Z*, and the ionic charge, for *s* and *p* block ions only, up to $Z = 36$
22. know that elements can be classified as *s*, *p* and *d*-block elements
23. understand that electronic configuration determines the chemical properties of an element
24. understand periodicity in terms of a repeating pattern across different periods
25. understand reasons for the trends in the following properties of the elements from periods 2 and 3 of the Periodic Table:
 - i the melting and boiling temperatures of the elements, based on given data, in terms of structure and bonding
 - ii ionisation energy based on given data or recall of the plots of ionisation energy versus atomic number
26. be able to illustrate periodicity using data, including electronic configurations, atomic radii, melting and boiling temperatures and first ionisation energies